

Quadratic differentials and conformal invariants

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Abstract We define a notion of conformal invariance associated with nested domains, suitable for characterizing higher-order information about mapping functions. We give an exposition of our results which yield an infinite-dimensional family of conformal invariants for nested hyperbolic simply-connected domains. Each invariant is specified by a quadratic differential which is admissible for the outer domain, and is strictly negative unless the inner domain is the outer domain minus trajectories of the quadratic differential. These invariants are furthermore monotonic. Using the aforementioned invariants, we show that one can obtain various classical estimates for bounded univalent functions, and in many cases extend them, by choosing particular quadratic differentials. We also explain the principles behind these results and their context within the literature.

Keywords Quadratic differential · Bounded univalent functions · Conformal invariants · Extremal problems

Mathematics Subject Classification 30C35 · 30C50 · 30C55 · 30C75 · 30F30

Dedicated to David Minda on the occasion of his retirement.

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1 A family of conformal invariants

1.1 Preliminaries

We will use the following conventions. The unit disk is denoted by $\mathbb{D} = \{z : |z| < 1\}$ and $\mathbb{C}$ denotes the complex plane.

A quadratic differential on a domain Ω is a $(2, 0)$ -differential $Q(z)dz^2$ on Ω where $Q(z)$ is a meromorphic function (see e.g. Lehto 1987). A curve $\gamma(t)$ on an interval (a, b) is called a trajectory of $Q(z)dz^2$ if $Q(\gamma(t))\gamma'(t)^2 \leq 0$ for all t in (a, b) . Endpoints may also be included if the curve extends continuously to $[a, b]$ and Q is non-singular on those endpoints. The pull-back of a quadratic differential $Q(z)dz^2$ on Ω under a locally one-to-one holomorphic function $f : D \rightarrow \Omega$ is the quadratic differential

$$f^*(Q(z)dz^2) = Q(f(w))f'(w)^2dw^2$$

on D . We say that a domain Ω is admissible for a quadratic differential $Q(z)dz^2$ if its boundary consists of a network of trajectories of $Q(z)dz^2$. We also say that $Q(z)dz^2$ is admissible for Ω in this case.

By the hyperbolic metric on a domain $\Omega \subset \mathbb{C}$ (if it exists) we mean the unique complete, constant curvature -4 conformal metric $\lambda_\Omega(z)^2|dz|^2$. For example,

$$\lambda_{\mathbb{D}}(z) = 1/(1 - |z|^2). \quad (1.1)$$

The hyperbolic metric is conformally invariant in the sense that if $g : D \rightarrow \Omega$ is a conformal bijection then

$$\lambda_\Omega(g(z))|g'(z)| = \lambda_D(z). \quad (1.2)$$

Finally, we denote the Green function of a domain Ω (if it exists) by $G_\Omega(z, w)$.

1.2 A notion of conformal invariance

In this section we define a certain notion of conformal invariance, and illustrate it with examples. General results will be given in Sect. 1.3 ahead. We will restrict to planar domains, although this notion of conformal invariance extends naturally to Riemann surfaces. See Schippers (2003a, b) and the review article of Schippers and Staubach (2016).

Let D_1 and D_2 be planar domains, and assume that $D_2 \subseteq D_1$. We might also specify an ordered set of points $z_1, \dots, z_n \in \text{cl}D_1$. A conformal invariant is a function I of such configurations $(D_1, D_2, z_1, \dots, z_n)$ satisfying

$$I(D_1, D_2, z_1, \dots, z_n) = I(g(D_1), g(D_2), g(z_1), \dots, g(z_n)) \quad (1.3)$$

whenever $g : D_1 \rightarrow E_1$ is a conformal bijection.

We will assume for convenience that the boundary of both domains is piecewise analytic, and the boundary of D_1 is in addition a Jordan curve. Note that with this assumption a conformal map g extends to the closure of D_1 , so that (1.3) makes

sense. This can be generalized to arbitrary domains conformal to the disk by use of the ideal boundary of D_1 (Schippers 2016).

Here are some examples.

Example 1.1 (Hyperbolic reduced module) For a hyperbolic planar domain define

$$I_1(D_1, D_2, z) = \log \left(\frac{\lambda_{D_1}(z)}{\lambda_{D_2}(z)} \right).$$

If $g : D_1 \rightarrow E_1$ is a conformal bijection, $g(D_2) = E_2$ and $g(z) = w$, using (1.2) we see that

$$I_1(D_1, D_2, z) = \log \left(\frac{\lambda_{D_1}(z)}{\lambda_{D_2}(z)} \right) = \log \left(\frac{\lambda_{E_1}(g(z))|g'(z)|}{\lambda_{E_2}(g(z))|g'(z)|} \right) = I_1(E_1, E_2, w).$$

Thus I_1 is a conformal invariant. $\square$

This seems to be due to Ahlfors and Beurling (1950) (formulated rather differently). There are many different characterizations of this invariant in the simply connected case, e.g. in terms of extremal metrics or Dirichlet energy (see Example 1.2 below). An elegant characterization was given by Barnard et al. (2005), who derived this quantity as a reduced module with respect to the hyperbolic metric of the outside domain. They restrict their attention to the disk, but their definition easily applies to the general case. For reduced modules on general Riemann surfaces see Minda (1970, 1972).

Example 1.2 For domains $D_2 \subseteq D_1$ possessing Green's functions G_{D_i} , and points $z, w \in D_2$, set

$$I_0(D_1, D_2, z, w) = G_{D_1}(z, w) - G_{D_2}(z, w)$$

since both Green's functions are conformally invariant, this is immediately seen to be a conformal invariant in the sense of Eq. (1.3).

Taking the limit as $w \rightarrow z$, in the simply connected case we recover the invariant of Example 1.1:

$$\lim_{w \rightarrow z} I_0(D_1, D_2, z, w) = -I_1(D_1, D_2, z).$$

$\square$

Example 1.3 For domains $D_2 \subseteq D_1$ with hyperbolic metrics $\lambda_{D_i}(z)^2 |dz|^2$, define the Christoffel symbols Γ_{D_i} by

$$\Gamma_{D_i}(z) = 2 \frac{\partial}{\partial z} \log \lambda_{D_i}(z). \quad (1.4)$$

For a conformal bijection $g : D_1 \rightarrow E_1$ it follow from (1.2) that

$$\Gamma_{D_i}(z) = \Gamma_{E_i}(g(z))g'(z) + \frac{g''(z)}{g'(z)}.$$

Thus

$$\Gamma_{D_1}(z) - \Gamma_{D_2}(z) = [\Gamma_{E_1}(g(z)) - \Gamma_{E_2}(g(z))]g'(z). \quad (1.5)$$

Assume now that D_1 and D_2 are simply connected. Fix a point p in ∂D_1 . For any point $z \in D_1$ let $\gamma'_{z,p}(0)$ be the tangent vector at z of the unique hyperbolic geodesic $\gamma_{z,p}(t)$ such that $\gamma_{z,p}(0) = z$ and whose limit point is p as $t \rightarrow \infty$. By conformal invariance of the hyperbolic metric we have that

$$\gamma'_{g(z),g(p)}(0) = g'(z)\gamma'_{z,p}(0). \quad (1.6)$$

Combining (1.5) and (1.6) we see that

$$I_2(D_1, D_2, z, p) = \left(\Gamma_{\lambda_{D_1}}(z) - \Gamma_{\lambda_{D_2}}(z) \right) \gamma'_{z,p}(0)$$

is a conformal invariant in the sense of (1.3). $\square$

Further examples of such conformal invariants can be found in Schippers (2003a, b) and Schippers and Staubach (2016).

By conformal invariance, we can choose $D_1 = \mathbb{D}$ and $D_2 = f(\mathbb{D})$ for some conformal bijection f . For Example 1.1 we then have the explicit expression

$$I_1(\mathbb{D}, f(\mathbb{D}), f(z)) = \log \frac{(1 - |z|^2)|f'(z)|}{(1 - |f(z)|^2)}. \quad (1.7)$$

For Example 1.3 we can find an explicit expression as follows. For $|p| = 1$, setting $\bar{q} = (p - z)/(1 - \bar{z}p)$ we have that the disk automorphism

$$T(w) = \bar{q} \frac{w + zq}{1 + \bar{z}qw}$$

takes 0 to z and 1 to p . Thus the curve $T(t)$ has the same tangent vector as $\gamma_{z,p}(t)$ at $t = 0$. From this we can compute that

$$\gamma'_{z,p}(0) = \frac{p - z}{1 - \bar{z}p} (1 - |z|^2). \quad (1.8)$$

Now by (1.2) we have

$$\lambda_{f(\mathbb{D})}(z) = \frac{1}{(1 - |f^{-1}(z)|^2)|f'(f^{-1}(z))|}.$$

Differentiating this and combining with (1.8) we arrive at

$$I_2(\mathbb{D}, f(\mathbb{D}), z, p) = \left[(1 - |z|^2) \frac{f''(f^{-1}(z))}{(f'(f^{-1}(z)))^2} + 2\bar{z} - \frac{(1 - |z|^2)\overline{f^{-1}(z)}}{(1 - |f^{-1}(z)|^2)f'(f^{-1}(z))} \right] \frac{p - z}{1 - \bar{z}p}. \quad (1.9)$$

One recognizes in these expressions the Minda-Peschl invariant derivatives, in this case with hyperbolic metric on the domain and range (see e.g. Ma and Minda 1997; in the expression for I_2 one must first compose with $f(z)$). They appear because invariance under disk automorphisms follows from conformal invariance. However, conformal invariance is a stronger condition. For example, the invariance of Green's function under an automorphism of a fixed domain is a weaker condition than the invariance of Green's function under conformal maps (Example 1.2).

Observe that I_1 involves the first derivative of f and I_2 involves the second derivative. We call the highest order of derivative of the conformal map f appearing in the explicit expression for the conformal invariant the “order” of the invariant. Conformal invariants of higher order than two can be found in Schippers (2003a, b).

We now ask the following deceptively simple question.

Question: How can one construct conformal invariants which involve arbitrary orders of derivatives of the mapping function?

Since the answer must involve at least a pair of domains, we are led directly to the notion of conformal invariance specified by (1.3).

We answer this question in the case of hyperbolic simply connected domains in the next section. Estimates are automatically produced by the construction. The answer also sheds light on the relation between functionals and quadratic differentials.

1.3 Definition of conformal invariants and the main theorems

We sketch the definition of the conformal invariants and state the main theorems. Careful definitions and rigorous proofs can be found in Schippers (2016).

Let D_1 and D_2 be a pair of hyperbolic simply connected domains in the plane, with $D_2 \subseteq D_1$. We assume D_2 is bounded by a network of analytic arcs which are trajectories of some quadratic differential $Q(z)dz^2$. This may seem restrictive, but it includes for example any domain bounded by an analytic Jordan curve. We also assume for ease of presentation that D_1 is bounded by an analytic Jordan curve. In general no restrictions on D_1 are necessary beyond that it is hyperbolic and simply connected (Schippers 2016). *These conditions will be in force through the rest of the paper unless otherwise specified.*

We consider triples $(D_1, D_2, Q(z)dz^2)$ such that

1. $Q(z)dz^2$ is admissible for D_1 ;
2. $D_2 \subseteq D_1$;
3. all poles of $Q(z)dz^2$ are in D_2 .

For each such triple, we define a conformal invariant. The definition is somewhat involved, if one demands a rigorous treatment. If these demands are relaxed, the

definition is easily stated. Given a triple $(D_1, D_2, Q(z)dz^2)$, let $\pi: \tilde{D}_1 \rightarrow D_1$ be a double covering with branch points of order two at each odd-order pole and zero of $Q(z)dz^2$. The pull-back $\pi^*(Q(z)dz^2)$ of the quadratic differential has a well-defined square root which is a one-differential (Schippers 2016, see also Minda and Rodin 1975). We then define singular multi-valued functions

$$x_1 = \int \sqrt{Q(\pi(z))\pi'(z)^2} dz \quad \text{and} \quad q_1 = \operatorname{Re} x_1. \quad (1.10)$$

It can be shown (Schippers 2016, Theorem 3.3) that q_1 is single-valued on an open neighbourhood of $D_1 \setminus D_2$. On the domain D_2 , we associate another singular harmonic function determined by the following Dirichlet problem: let u be the unique function on $\tilde{D}_2$ (where $\tilde{D}_2$ is the subset of $\tilde{D}_1$ covering D_1) such that

$$u(z) = q_1(z) \quad \text{for } z \in \partial\tilde{D}_2 \quad \text{and} \quad \Delta u(z) = 0 \quad \text{for } z \in \tilde{D}_2,$$

where Δ is the Laplacian on $\tilde{D}_2$. We then set

$$q_2 = q_1 + u$$

in $\tilde{D}_2$. Define

$$m(D_1, D_2, Q(z)dz^2) = \int_{\partial\tilde{D}_2} q_2 * dq_2 \quad (1.11)$$

where $*dq_2$ is the Hodge star operator on $\tilde{D}_2$. In terms of a local parameter on $\tilde{D}_1$, we can write the integral (1.11) locally along a sub-arc γ of $\partial\tilde{D}_2$ as

$$\int_{\gamma} q_2 \frac{\partial q_1}{\partial n} ds,$$

where ds is arc length and n is the unit outward normal. The quantity $m(D_1, D_2, Q(z)dz^2)$ is independent of the various choices made, for example the double cover and sign of q_1 (Schippers 2016).

The importance of this quantity is that it is conformally invariant.

Theorem 1.1 (Conformal invariance, Schippers 2016) *If $g: E_1 \rightarrow D_1$ is a conformal bijection and $g(E_2) = D_2$ then*

$$m(E_1, E_2, g^*Q(z)dz^2) = m(D_1, D_2, Q(z)dz^2).$$

That is, $m(D_1, D_2, Q(z)dz^2)$ is a conformal invariant.

This can also be viewed as an invariant of the form (1.3). To see this, observe that the quadratic differential $Q(z)dz^2$ is uniquely determined by the locations and orders of its poles and zeros, together with the assumption that the boundary of D_1 is a trajectory of $Q(z)dz^2$. Thus $m(D_1, D_2, Q(z)dz^2)$ is a particular function of the location of zeros and poles, given by choosing their orders. Since pull-back preserves the order of poles and zeros we see that $m(D_1, D_2, Q(z)dz^2)$ is conformally invariant in the sense of (1.3).

Furthermore, judiciously applying Green's identities and positivity of Dirichlet energy, it follows that

Theorem 1.2 (Negativity, Schippers 2016) *For all $D_2 \subseteq D_1$, $m(D_1, D_2, Q(z)dz^2) \leq 0$ with equality if and only if D_2 is D_1 minus trajectories of $Q(z)dz^2$.*

The conformal invariant m is also monotonic. Explicitly, the harmonic function q_2 induces a quadratic differential on D_2 , namely

$$Q_2(z)dz^2 = 4 \left(\frac{\partial q_2 \circ \pi^{-1}}{\partial z} \right) dz^2$$

where $\pi : \tilde{D}_1 \rightarrow D_1$ is the covering map. Call $Q_2(z)dz^2$ the quadratic differential on D_2 induced by $Q(z)dz^2$. We then have

Theorem 1.3 (Monotonicity, Schippers 2016) *Let $D_2 \subseteq D_1$ be simple domains and $Q(z)dz^2$ be a quadratic differential admissible for D_1 . Let $Q_2(z)dz^2$ be the quadratic differential on D_2 induced by $Q(z)dz^2$. If $D_3 \subseteq D_2$ is a simple domain, then*

$$m(D_1, D_3, Q(z)dz^2) - m(D_1, D_2, Q(z)dz^2) = m(D_2, D_3, Q_2(z)dz^2).$$

In particular, $m(D_1, D_3, Q(z)dz^2) \leq m(D_1, D_2, Q(z)dz^2)$.

1.4 Relation to the literature

Conformally invariant methods for producing estimates on domain functions and mapping functions are of course ubiquitous in the literature. There are for example potential theoretic methods involving capacitance or harmonic measure, as in Baernstein and Solynin (2013), Dubinin (2014), or Duren and Pfaltzgraff (1999); extremal length, as in Minda (1970, 1972), Minda and Rodin (1975), and the monographs of Jenkins (1958), Kuz'mina (1980) or Vasil'ev (2002); or Dirichlet energy methods as in Dubinin (2014). Needless to say this sample can hardly represent the vast literature. Furthermore, the line between these methods is at times artificial, because of the many theorems relating conformal invariants, see e.g. Ahlfors (1973).

Nehari's innovation was to adapt Dirichlet energy methods to allow singular functions (Nehari 1953). This key innovation allowed higher order estimates, as will be explained in this paper. In this paper and Schippers (2016), by identifying the role of quadratic differentials in Nehari's method, we generalize it significantly, as well as giving an elegant conformally invariant formulation. This is a manifestation of the so-called Teichmüller principle, as we discuss in Sect. 3.1. The sense in which our results generalize those of Nehari is explained in Sect. 3.2.

The appearance of quadratic differentials here suggests a relation with the method of quadratic differentials/extremal metrics of Jenkins (1958) (see also Pommerenke 1975, Chapter 8). This is based on the length-area method, which has roots in ideas of Teichmüller (1938), Grötzsch (1928a, b), and others. These methods associate higher-order functionals, and estimates on them, to quadratic

differentials. However, the resulting functionals are not manifestly conformally invariant. In principle, one might attempt to construct reduced modules by removing domains and taking limits of modules of curve families. However, in practice it is difficult to maintain conformal invariance after the limiting process. At present, this has only been accomplished (even implicitly) in the literature for quadratic differentials with poles of order two or lower, which are associated with invariants of order one (see e.g. Example 2.1, Dubinin 2014, Chapter 2 and Minda and Rodin 1975).

Although the approach in this paper is altogether different from the extremal metrics approach, there ought to be a connection. By the Beurling theorem (Ahlfors 1973), the Dirichlet energy of solutions of mixed boundary value problems are equal to the modules of associated curve families. Thus we conjecture a relation between the conformal invariants here and extremal length. However, again one finds that the presence of singularities in the Nehari technique, and the associated limiting procedure for reduced modules, makes it unclear how to put this idea into practice. In some sense, an explicit relation would extend the Beurling theorem.

The relation with Schiffer's variational theorems is briefly discussed in Sect. 3.3. More generally, we observe that the notion of conformal invariance defined here has connections to Schiffer's comparison theory of domains (Schiffer 1977, Appendix); see also Schippers and Staubach (2016). We were also inspired many years ago by the "intrinsic" formulations of two-point distortion theorems of Kim and Minda (1994) and Ma and Minda (1997).

2 Applications to function theory

2.1 Examples

We will apply the results of the previous section to deduce inequalities for bounded univalent maps. Various inequalities can be obtained simply by choosing quadratic differentials. The purpose of this section is to demonstrate the unity and very general nature of Theorems 1.2 and 1.3. To do so, we reproduce several known results (though we mention a new inequality from Schippers (2016) and a few new monotonicity theorems).

Remark 2.1 Although the monotonicity and positivity theorem generates infinitely many inequalities to explore, the production of such inequalities is a dull end goal. It is much more interesting to understand (for example) the relation between quadratic differentials, functionals on classes of conformal mappings, and boundary points of classes of conformal mappings. Aside from well-known heuristic principles, this is still poorly understood, and there are many open questions. See for example Schaeffer and Spencer (1950) and Duren (1983, Chapter 11); see also Pfluger's intriguing results for the case of the third coefficient body (Pfluger 1988). The estimates below are intended to illustrate Theorems 1.1, 1.2, and 1.3.

We will need an alternative expression for the conformal invariants, which facilitates specific applications. For domains $D_2 \subseteq D_1$ and a quadratic differential $Q(z)dz^2$ satisfying properties (1) to (3), let x_1 be the primitive (1.10) of $\sqrt{\pi^*(Q(z)dz^2)}$ on the double cover $\tilde{D}_1$ of D_1 . Let q_2 be the harmonic function on $\tilde{D}_2$ such that $q_2 - \operatorname{Re}(x_1)$ is harmonic on $\tilde{D}_2$ and let x_2 be the multi-valued analytic function on $\tilde{D}_2$ such that $\operatorname{Re}(x_2) = q_2$. It can be shown (Schippers 2016) that the conformal invariant is equal to the following well-defined contour integral:

$$m(D_1, D_2, Q(z)dz^2) = \operatorname{Re} \left(\frac{1}{i} \int_{\partial \tilde{D}_2} (x_1 - x_2) \frac{\partial x_1}{\partial z} dz \right). \quad (2.1)$$

By conformal invariance, one can assume without loss of generality that D_1 is the disk $\mathbb{D}$ and D_2 is the image of $\mathbb{D}$ under a conformal map. We do this in several examples below.

Example 2.1 (Hyperbolic reduced module) In this section we show that one can obtain the hyperbolic reduced module for a domain inside a conformal disk, by choosing a quadratic differential with a single double pole.

For $a \in \mathbb{D}$, set

$$Q(z)dz^2 = \frac{(1 - |a|^2)}{(1 - \bar{a}z)^2(z - a)^2} dz^2$$

which, up to a real scale factor, is the unique quadratic differential admissible for $\mathbb{D}$ with a double pole at a . Since this is a perfect square, we can find the function x on $\mathbb{D}$ directly and do not need to pass to the double cover. We immediately have that

$$x_1(z) = \log \left(\frac{z - a}{1 - \bar{a}z} \right)$$

(note $\operatorname{Re}(x_1)$ is verifiably single-valued on $\mathbb{D} \setminus \{a\}$). If $\mathbb{D} = F(D_2)$ for a conformal bijection F we have

$$x_2(z) = \log \left(\frac{F(z) - F(a)}{1 - \overline{F(a)}F(z)} \right).$$

By Eq. (2.1) we have

$$\begin{aligned} m(\mathbb{D}, D_2, Q(z)dz^2) &= 4\pi \operatorname{Re} \left(\frac{1}{2\pi i} \int_{\partial D_1} \log \left(\frac{z - a}{F(z) - F(a)} \cdot \frac{1 - \overline{F(a)}F(z)}{1 - \bar{a}z} \right) \right. \\ &\quad \times \left. \left(\frac{1}{z - a} + \frac{\bar{a}}{1 - \bar{a}z} \right) dz \right) \\ &= 4\pi \log \frac{(1 - |F(a)|^2)}{(1 - |a|^2)|F'(a)|}. \end{aligned}$$

If $\lambda_{D_2}(a) = |F'(a)|/(1 - |F(a)|^2)$ denotes the hyperbolic density of D_1 and similarly $\lambda_{\mathbb{D}}(a) = 1/(1 - |a|^2)$, we see that

$$m(\mathbb{D}, D_1, Q(z)dz^2) = 4\pi \log \frac{\lambda_{\mathbb{D}}(a)}{\lambda_{D_2}(a)}.$$

The negativity and monotonicity of this module, guaranteed by Theorems 1.2 and 1.3, in this case are equivalent to the Schwarz lemma. Since $Q(z)dz^2$ has no zeros on the boundary, there are no trajectories of $Q(z)dz^2$ meeting the boundary, and so equality holds if and only if $D_2 = \mathbb{D}$.

By conformal invariance, we have that if $Q(z)dz^2$ is the unique quadratic differential which is admissible for D_1 , with a single pole of order two at w and no other zeros or poles, then

$$m(D_1, D_2, Q(z)dz^2) = 4\pi \log \frac{\lambda_{D_1}(w)}{\lambda_{D_2}(w)} = 4\pi I_1(D_1, D_2, w).$$

□

Example 2.2 (Schippers 2016) Let

$$Q(z)dz^2 = -e^{-i\psi} \frac{(e^{i\psi} + z)^2}{z^2(z-r)(z-1/r)} dz^2 \quad (2.2)$$

for $0 < r < 1$. It is easily checked that this is admissible for $D_1 = \mathbb{D}$. Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be a one-to-one map such that $f(0) = 0$, and set $D_2 = f(\mathbb{D})$. With this choice of quadratic differential we have

$$\begin{aligned} m(\mathbb{D}, D_2, Q(z)dz^2) &= 4\pi \operatorname{Re} \left[-e^{i\psi} \log \left(\frac{wf'(0)}{f(w)} \frac{1 - |f(w)|^2}{1 - |w|^2} \right) \right] \\ &\quad + \log \left(\frac{1 + |f(w)|}{1 - |f(w)|} \frac{1 - |w|}{1 + |w|} \right) \\ &\leq 0 \end{aligned} \quad (2.3)$$

where $w = f^{-1}(r)$. Here the first term uses the unique branch of logarithm such that $\log(wf'(0)/f(w)) \rightarrow 0$ as $w \rightarrow 1$ and the second uses the principal branch. The upper bound is a consequence of Theorem 1.2, combined with the fact that the expression $wf'(0)/f(w)$ is invariant under a rotation $e^{-i\theta}f(e^{i\theta}z)$. Note that the expression is also monotonic in the sense that the right hand side of the top row is greater for g such that $f(\mathbb{D}) \subseteq g(\mathbb{D})$ and $f(w) = g(w)$.

Choosing $e^{i\psi} = \pm 1$, we obtain the Pick growth theorems (Pick 1917). The monotonicity property of the Pick growth theorems does not seem to have been observed. For $e^{i\psi} = \pm i$, we obtain argument estimates for bounded univalent functions. Argument estimates were also obtained by Pommerenke and Vasil'ev (2000) which are sharp for maps onto a radially slit domain. The argument estimate here is distinct, since by Theorem 1.2 equality in (2.3) holds for maps onto $\mathbb{D}$ minus a trajectory of (2.2), which for $e^{i\psi} = \pm i$ is not a radial slit.

Note that the other values of $e^{i\psi}$, corresponding to the locations of the zero of $Q(z)dz^2$, produce inequalities interpolating the lower and upper bound in the Pick growth theorem. $\square$

Example 2.3 (Schippers 2016) In the previous example, we fixed one of the poles at 0; however, by conformal invariance we may place them anywhere. If the zero lies on the terminal point of the hyperbolic geodesic through these poles, we get a form of the Pick growth theorems in terms of the hyperbolic metric. These are due to Ma and Minda (1997) (in a different but algebraically equivalent form).

Let $Q_\tau(z)dz^2$ be the unique quadratic differential admissible for D_1 with a double pole at $u \in \mathbb{D}$, a simple pole at $v \in \mathbb{D}$, and a double zero at the terminal point τ of the hyperbolic geodesic ray starting at u and passing through v ; assume there are no other zeros or poles. We then have

$$m(D_1, D_2, Q_\tau(z)dz^2) = \log \left(\frac{e^{-4d_{D_2}(u,v)} - 1}{e^{-4d_{D_1}(u,v)} - 1} \cdot \frac{\lambda_{D_1}(u)}{\lambda_{D_2}(u)} \right) \leq 0 \quad (2.4)$$

where d_{D_i} denotes the hyperbolic distance.

Now let ρ be the other terminal point of the geodesic through u and v (that is, the terminal point of the geodesic ray starting at v and passing through u). Then

$$m(D_1, D_2, Q_\rho(z)dz^2) = \log \left(\frac{e^{4d_{D_1}(u,v)} - 1}{e^{4d_{D_2}(u,v)} - 1} \cdot \frac{\lambda_{D_2}(u)}{\lambda_{D_1}(u)} \right) \leq 0. \quad (2.5)$$

The Negativity Theorem 1.2 applied to (2.4) is equivalent to the lower bound in the two-point distortion theorem of Ma and Minda (1997); applying it to (2.5) results in the upper bound. Note that Ma and Minda's distortion theorem is conformally invariant after a minor adjustment, so Theorem 1.2 does not give new information. On the other hand, Theorem 1.3 implies the monotonicity of both quantities (fixing u and v and varying the inner domain D_2), which appears to be a new result. $\square$

Example 2.4 (Second order) We consider a quadratic differential admissible for $\mathbb{D}$ with a third-order pole at 0. Multiplying Q by a multiplicative constant changes x by a multiplicative constant, and hence the same is true for x_2 for $D_2 \subset \mathbb{D}$. Thus the module $m(\mathbb{D}, D_2, Q(z)dz^2)$ will change only by a multiplicative constant. If the boundary of the disk is a trajectory, up to a multiplicative constant we can assume that Q has the form

$$Q(z) = \frac{1}{z^2} \left(\frac{e^{-i\theta}}{z} + \beta + e^{i\theta}z \right)$$

for some $\beta \geq 2$ (if $\beta < 2$, then $\mathbb{D}$ is not admissible for $Q(z)dz^2$). If $\beta > 2$, then Q has no zeros on $\partial\mathbb{D}$, so the upper bound of the resulting monotonic functional will be attained only by the identity function. We restrict ourselves to the case that $\beta = 2$, in which case we have

$$Q(z) = \frac{1}{z^3} \left(e^{i\theta/2}z + e^{-i\theta/2} \right)^2. \quad (2.6)$$

Let $\tilde{D}$ be a double cover of $\mathbb{D}$ with a branch point of order two at 0. Explicitly, we may choose $\pi : \tilde{D} \rightarrow \mathbb{D}$ where $\pi(w) = w^2$ and $\tilde{D} = \mathbb{D}$. We now compute x_1 and x_2 . Observe that $\pi^*(Q(z)dz^2) = 4w^2Q(w^2)dw^2$. Setting $a = e^{i\theta/2}$,

$$\begin{aligned} x'_1(w) &= 2\sqrt{w^2Q(w^2)} = 2\left(\frac{1}{w^4}\left(aw^2 + \frac{1}{a}\right)^2\right)^{1/2} \\ &= 2a + \frac{2}{aw^2}, \end{aligned} \quad (2.7)$$

where we have made an arbitrary choice of the sign of the square root. Thus

$$x_1(w) = 2aw - \frac{2}{aw}. \quad (2.8)$$

It is easily verified that the real part of x_1 is zero for $|w| = 1$, and that x_1 is single-valued.

Let $\tilde{F}(w) = c_1w + c_3w^3 + \dots$ be a map $\tilde{F}(w) = \sqrt{F(w^2)}$ from $\tilde{D}$ onto $\tilde{D}_2 = \pi^{-1}(D_2)$, where $F : D_2 \rightarrow \mathbb{D}$ is a conformal bijection satisfying $F(0) = 0$. We claim that

$$x_2(w) = -2\frac{\tilde{F}'(0)}{a\tilde{F}(w)} + 2a\overline{\tilde{F}'(0)}\tilde{F}(w). \quad (2.9)$$

Since $\overline{\tilde{F}(w)} = 1/\tilde{F}(w)$ on $\partial\tilde{D}_2$, $\operatorname{Re} x_2(w) = 0$ on $\partial\tilde{D}_2$. Computing the Laurent series of x_2

$$x_2(w) = -\frac{2}{aw} + \left(\frac{2c_3}{ac_1} + 2a|c_1|^2\right)w + O(w^2), \quad (2.10)$$

we see that $x_1 - x_2$ is non-singular. This proves the claim.

Now we can apply (2.1) to evaluate $m(\mathbb{D}, D_2, Q(z)dz^2)$. We have by (2.1) and (2.7) that

$$\begin{aligned} m(\mathbb{D}, D_2, Q(z)dz^2) &= \operatorname{Re} \left(\frac{1}{i} \int_{\partial\tilde{D}_2} (x_1(w) - x_2(w)) \cdot x'_1(z) dz \right) \\ &= 4\pi \operatorname{Re} \left(\frac{1}{a} (x_1 - x_2)'(0) \right). \end{aligned}$$

Thus

$$m(\mathbb{D}, D_2, Q(z)dz^2) = 8\pi \operatorname{Re} \left(1 - |c_1|^2 - a^{-2}\frac{c_3}{c_1} \right). \quad (2.11)$$

Letting $f : \mathbb{D} \rightarrow D_2$ with power series $f(z) = a_1z + a_2z^2 + \dots$ be a holomorphic bijection such that $f(0) = 0$, and setting $F = f^{-1}$, we have, using $|c_1|^2 = 1/|a_1|$ and $c_3/c_1 = -a_2/2a_1^2$ that

$$m(\mathbb{D}, f(\mathbb{D}), Q(z)dz^2) = 4\pi \operatorname{Re} \left(2 - \frac{2}{|a_1|} + e^{-i\theta} \frac{a_2}{a_1^2} \right). \quad (2.12)$$

which implies the following result.

Theorem 2.2 *Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be a one-to-one holomorphic map such that $f(0) = 0$, with power series expansion $f(z) = a_1z + a_2z^2 + \cdots$. Set*

$$\Phi(f) = \operatorname{Re} \left[e^{-i\theta} \frac{a_2}{a_1^2} - 2 \left(\frac{1}{|a_1|} - 1 \right) \right].$$

Then $\Phi(f) \leq 0$ with equality if and only if f maps onto a disk minus trajectories of $Q(z)dz^2$. Furthermore $\Phi(f)$ is monotonic in the sense that if $f_1(\mathbb{D}) \subseteq f_2(\mathbb{D})$ then $\Phi(f_1) \leq \Phi(f_2)$.

Proof For $Q(z)dz^2$ given by (2.6), if $f(\mathbb{D})$ is simple then $m(\mathbb{D}, f(\mathbb{D}), Q(z)dz^2) = 8\pi\Psi(f)$. Thus the claim follows from Theorems 1.2 and 1.3. Note that although the invariants were only defined for D_2 bounded by analytic curves, one can apply a density argument to extend $\Phi(f)$ and the result to arbitrary injective holomorphic maps $f : \mathbb{D} \rightarrow \mathbb{D}$. $\square$

The inequality $\Phi(f) \leq 0$ is just the familiar second coefficient estimate for bounded univalent functions (Tammi 1978, p. 8). The monotonicity of this expression does not seem to have been observed.

Remark 2.3 In the computation above one recognizes the familiar use of the square root transform $\sqrt{F(w^2)}$ in deriving the second coefficient estimate. Indeed the branched double cover is a systematization and considerable generalization of the square root transform technique.

Example 2.5 (*Expressing the previous example in terms of hyperbolic metrics*) Again set $D_1 = \mathbb{D}$ and $D_2 = f(\mathbb{D})$ for a one-to-one holomorphic map $f : \mathbb{D} \rightarrow \mathbb{D}$ such that $f(0) = 0$. We express the previous example in terms of the hyperbolic metric. Given a domain D_1 let $g : D_1 \rightarrow \mathbb{D}$ be the unique conformal bijection such that $g(a) = 0$ and $g(p) = -e^{-i\psi}$. Set $Q_{a,p}(z)dz^2 = g^*(Q(z)dz^2)$ where $Q(z)dz^2$ is the differential (2.6) on $\mathbb{D}$. Observe that $Q_{a,p}(z)dz^2$ is the unique admissible quadratic differential on D_1 with a triple pole at a and a simple zero at $p \in \partial\mathbb{D}$, up to a positive multiplicative factor. Setting $z = 0$ and $p = -e^{-i\psi}$ in (1.9) and $z = 0$ in (1.7) we obtain

$$\frac{1}{2\pi} m(\mathbb{D}, f(\mathbb{D}), Q(z)dz^2) = 4 - 4 \exp[-I_1(\mathbb{D}, f(\mathbb{D}), 0)] - \operatorname{Re} I_2(\mathbb{D}, f(\mathbb{D}), 0, -e^{-i\psi}).$$

Thus by Theorem 1.1 we see that

$$\frac{1}{2\pi} m(D_1, D_2, Q_{a,p}(z)dz^2) = 4 - 4 \exp[-I_1(D_1, D_2, a)] - \operatorname{Re} I_2(D_1, D_2, a, p).$$

In general, obtaining expressions in terms of the hyperbolic metric becomes more cumbersome as the order of the estimate increases. $\square$

Example 2.6 (Third order) It is not hard to verify that the most general quadratic differential admissible for the disk, with a fourth order pole at 0 and no other poles, and two distinct double zeroes on the boundary is, up to a positive multiplicative constant,

$$Q(z)dz^2 = \frac{1}{z^4} e^{-i(\psi+\phi)} (z + e^{i\psi})^2 (z + e^{i\phi})^2 dz^2. \quad (2.13)$$

This determines the monotonic conformal invariant up to a constant factor.

The resulting inequality below was first obtained by De Temple (1971) using a variational method. De Temple (1971) later gave another proof using the Loewner method. Jenkins (1960) earlier derived an unbounded version (with the outer domain $\mathbb{C}$) from the general coefficient theorem. In no case was the result shown to be conformally invariant or monotonic.

Now we compute the module $m(\mathbb{D}, D_2, Q(z)dz^2)$. To reduce clutter set

$$\alpha = e^{i(\psi+\phi)/2}, \quad \beta = 2 \cos(\phi - \psi)/2$$

and we see that we may write

$$x'_1(z) = \frac{e^{-i(\psi+\phi)/2}}{z^2} (z + e^{i\psi})(z + e^{i\phi}) = \frac{\alpha}{z^2} + \frac{\beta}{z} + \bar{\alpha}. \quad (2.14)$$

Since $Q(z)$ is a perfect square, there is no need to lift to the double cover. We compute

$$x_1(z) = -\frac{\alpha}{z} + \beta \log z + \bar{\alpha}z.$$

Here x_1 is multiple valued, although the real part is single valued on $\mathbb{D} \setminus \{0\}$ since β is real.

Now let D_2 be a simply connected domain in $\mathbb{D}$ containing 0. Let $F : D_2 \rightarrow \mathbb{D}$ be a conformal map such that $F(0) = 0$. We then have that

$$x_2(z) = -\alpha \frac{F'(0)}{F(z)} + \beta \log F(z) + \overline{\alpha F'(0)} F(z).$$

It is easily verified that the real part of x_2 is single-valued on $D_2 \setminus \{0\}$ and is zero on ∂D_2 ; furthermore $x_1 - x_2$ is non-singular on D_2 . If F has power series $F(z) = A_1 z + A_2 z^2 + A_3 z^3 + \dots$ then we can compute that

$$\begin{aligned} \alpha \left(\frac{F'(0)}{F(z)} - \frac{1}{z} \right) &= \alpha \left(-\frac{A_2}{A_1} + \left(-\frac{A_3}{A_1} + \frac{A_2^2}{A_1^2} \right) z \right) + O(z^2), \\ -\beta \log \frac{F(z)}{z} &= -\beta \log |A_1| - \beta it - \beta \frac{A_2}{A_1} z + O(z^2), \end{aligned}$$

where $t = \arg A_1$ depends on the choice of the branch of logarithm, and finally

$$\bar{\alpha} \left(z - \overline{F'(0)} F(z) \right) = \bar{\alpha} \left(1 - |A_1|^2 \right) z + O(z^2).$$

Thus

$$\begin{aligned}(x_1 - x_2)(z) &= \left[-\alpha \frac{A_2}{A_1} - \beta \log |A_1| - \beta i t \right] \\ &\quad + \left[\alpha \left(-\frac{A_3}{A_1} + \frac{A_2^2}{A_1^2} \right) - \beta \frac{A_2}{A_1} + \overline{\alpha} (1 - |A_1|^2) \right] z + O(z^2).\end{aligned}$$

Note that the real part of $x_1 - x_2$ is well-defined since β is real.

Using (2.1) we compute

$$\begin{aligned}m(\mathbb{D}, D_2, Q(z)dz^2) &= 2\operatorname{Re} \left[\frac{1}{i} \int_{\partial D_2} (x_1(z) - x_2(z)) x_1'(z) dz \right] \\ &= 4\pi \operatorname{Re} \left[\alpha^2 \left(-\frac{A_3}{A_1} + \frac{A_2^2}{A_1^2} \right) - 2\alpha\beta \frac{A_2}{A_1} - \beta^2 \log |A_1| + |\alpha|^2 (1 - |A_1|^2) \right].\end{aligned}$$

The factor of 2 after the first equality is due to the fact that in the definition of m we integrate over the boundary of the lift of D_2 to the cover, and thus double the value of the contour integral over ∂D_1 . Denoting $f(z) = F^{-1}(z) = a_1 z + a_2 z^2 + \dots$ and setting

$$\begin{aligned}\Psi(f) &= 4\pi \operatorname{Re} \left[\alpha^2 \left(\frac{a_3}{a_1^3} - \frac{a_2^2}{a_1^4} \right) + 2\alpha\beta \frac{a_2}{a_1^2} + \beta^2 \log |a_1| - \frac{|\alpha|^2}{|a_1|^2} (1 - |a_1|^2) \right] \\ &= 4\pi \operatorname{Re} \left[e^{i(\psi+\phi)} \left(\frac{a_3}{a_1^3} - \frac{a_2^2}{a_1^4} \right) + 4e^{i(\psi+\phi)/2} \cos(\phi/2 - \psi/2) \frac{a_2}{a_1^2} \right. \\ &\quad \left. + 4\cos^2(\phi/2 - \psi/2) \log |a_1| - \frac{1}{|a_1|^2} (1 - |a_1|^2) \right],\end{aligned}\tag{2.15}$$

then we have the following result.

Theorem 2.4 *Let $f: \mathbb{D} \rightarrow \mathbb{D}$ be a one-to-one holomorphic map such that $f(0) = 0$. Thus $\Psi(f) \leq 0$ with equality if and only if f maps onto $\mathbb{D}$ minus trajectories of $Q(z)dz^2$. Furthermore if $f_1(\mathbb{D}) \subseteq f_2(\mathbb{D})$ then $\Psi(f_1) \leq \Psi(f_2)$.*

Proof If $f(\mathbb{D})$ is simple then $m(\mathbb{D}, f(\mathbb{D}), Q(z)dz^2) = \Psi(f)$ where $Q(z)dz^2$ is given by (2.13). The claim now follows from Theorems 1.2 and 1.3, and a density argument. $\square$

The inequality of DeTemple (1971, Theorem 2.1, any of the following with preference for the first: (1) first case (2) First Case (3) case one (4) Case One) is equivalent to $\Psi(f) \leq 0$ if one chooses $\alpha = e^{i\theta} a_1 / |a_1|$ and $\beta = 2\sigma / |a_1|$, where σ thus ranges between 0 and $|a_1|$. Note that one must also take into account a change of notation for the coefficients of the power series. The monotonicity of Ψ has not been observed.

3 General remarks

3.1 Teichmüller's principle

Teichmüller's principle states that an estimate involving the $n + 1$ st derivative of a map at a point z is associated with a quadratic differential with a pole of order n at z . We may include as many points as we like. The name "Teichmüller's principle" appears to be due to Jenkins (1958). The reference to Teichmüller provided by Jenkins seems to contain no such statement, although clearly Teichmüller's theorem in Teichmüller (1938) is an example of the principle (see also Duren 1983). Teichmüller acknowledges the appearance of quadratic differentials in Schiffer's variational technique in the same paper (Teichmüller 1938, footnote).

The imprecision of the phrase "is associated with" does not lessen the mathematical interest of Teichmüller's principle. Teichmüller's theorem and Jenkins' general coefficient theorem (Jenkins 1958) are precise instances of the principle: if one is given a quadratic differential with a pole of order $n + 1$ these theorems produce a specific inequality involving the n th derivative of the mapping function. Schiffer's variational method is another: given a reasonably regular functional, the extremal functional must map onto the domain minus trajectories of a quadratic differential, whose poles have order one more than the order of the functional.

Theorems 1.1, 1.2, and 1.3 give yet another precise version of the principle. The reader may verify that Teichmüller's principle holds for the conformal invariants (1.11) in all the examples given in the previous section.

By examining (2.1) we can see that the conformal invariants (1.11) satisfy the principle in general. To see this, fix a pole of odd order $n + 1$ of the quadratic differential $Q(z)dz^2$ (the even case is much easier to analyze, and we leave it to the reader). By conformal invariance, we may assume that the pole is at 0. The covering map π will be branched at 0 and behaves as $\pi(w) \sim w^2$. Now let $F : D_2 \rightarrow \mathbb{D}$ be such that $F(0) = 0$, and let $\tilde{F}(w) = c_1w + c_2w^2 + \dots$ denote the lift of F to the double-cover. The claim now follows from the following argument:

1. $\pi^*(Q(z)dz^2) \sim 4w^2Q(w^2)dw^2$ will have a pole of order $2n$ at 0;
2. x'_1 thus has a pole of order n at 0, and therefore since $x_1 - x_2$ is non-singular, x_2 has a pole of order $n - 1$ at 0;
3. the highest coefficient of $\tilde{F}(w)$ appearing in the n th coefficient of $x_1 - x_2$ is produced by a term of the form $c/\tilde{F}(w)^{n-1}$ for some constant c (see e.g. Eq. (2.9));
4. thus the highest coefficient of $\tilde{F}$ in the contour integral (2.1) will appear in the $n - 1$ st term of the Laurent expansion of $1/\tilde{F}(w)^{n-1}$. This coefficient is easily computed to be c_{2n-1} ;
5. $\tilde{F}(w)$ behaves as $\sqrt{F(w^2)}$ near 0, so the coefficient c_{2n-1} of $\tilde{F}$ contains the n th coefficient of the Taylor series of $f = F^{-1}$, and no higher coefficients.

This proves Teichmüller's principle for the conformal invariants (1.11).

3.2 Relation to Nehari's method

Again, we restrict to the simply connected case; some remarks on generalizations follow. In this section we discuss how Nehari's method corresponds to the case that $Q(z)dz^2 = x'_1(z)dz^2$ where x_1 is a holomorphic function on D_1 with single-valued real part on $D_1 \setminus D_2$. Thus Theorems 1.2 and 1.3 generalize Nehari's method significantly. Extremal length methods (Jenkins 1958; Kuz'mina 1980; Vasil'ev 2002) do not require this restriction.

In Nehari (1953), associates inequalities to a harmonic singularity function, which he calls S . He makes no assumptions (beyond regularity) about the behaviour of S on the boundary of the outer domain D_1 . Given S , we let p_2 be the unique harmonic function such that $p_2 = 0$ on ∂D_2 and $S + p_2$ is harmonic on D_2 . Nehari's functional is defined to be

$$M_{\text{Nehari}}(D_1, D_2, S) = \int_{\partial D_2} S \frac{\partial p_2}{\partial n} ds. \quad (3.1)$$

Nehari showed that this is positive and monotonic.

To compare with the results presented here, let p_1 be the harmonic function on D_1 such that $p_1 = 0$ on ∂D_1 and $S + p_1 = 0$ on D_1 . Setting

$$Q(z)dz^2 = 4 \left(-\frac{\partial p_1}{\partial z} \right)^2 dz^2 = x'_1(z)^2 dz^2,$$

(where the latter is by the Cauchy-Riemann equations), we have that $Q(z)dz^2$ is admissible for D_1 and Schippers (2016)

$$-\frac{1}{2}m(D_1, D_2, Q(z)dz^2) = M_{\text{Nehari}}(D_1, D_2, S) - M_{\text{Nehari}}(D_1, D_1, S).$$

Thus we have

Proposition 3.1 (Schippers 2016) *For simply connected D_1 and D_2 , every Nehari functional M is of the form*

$$M = -\frac{1}{2}m(D_1, D_2, Q(z)dz^2) + C$$

with $Q(z) = 4(-\partial p_1/\partial z)^2$, where p_1 is some harmonic, singular function which is zero on ∂D_1 .

Thus, up to constants, Nehari's functionals are in one-to-one correspondence with those conformal invariants $m(D_1, D_2, Q(z)dz^2)$ such that $Q(z)dz^2 = x'_1(z)dz^2$ for some multi-valued analytic function x_1 with single-valued real part. It is also a striking fact that the constant term is the value of the Nehari functional when $D_2 = D_1$. That is, the functionals $m(D_1, D_2, x'_1(z)dz^2)$ are canonically in one-to-one correspondence with those Nehari functionals whose lower bound is zero. The relation between the admissibility of Nehari's singularity function and the extremal value of zero was first observed by the author in Schippers (2010).

Remark 3.2 I am grateful to Peter Duren for the following observation. In Schiffer's variational method, it is often advantageous to choose a functional so that the resulting quadratic differential is a perfect square; and in applications Schiffer himself often did this. This observation was one of our motivations for removing the restriction in Nehari's method.

Remark 3.3 The restriction to perfect squares is not present in the extremal metrics methods. If the quadratic differential is not a perfect square, then one applies a substitution to make it so. See for example the proofs of (Jenkins 1960, Theorem 1), (Kuz'mina 1980, Theorem 0.2), and (Vasil'ev 2002, Theorem 3.1.6). This is closely related to the so-called canonical map $\zeta = \int \sqrt{Q(z)} dz$ (Jenkins 1958, Chapter 3, Jenkins 1960, Theorem 1, Minda and Rodin 1975), under which the quadratic differential has the form $(d\zeta)^2$. In our generalization of Nehari's method, the function x_1 in (1.10) in some sense corresponds to the canonical map, and the substitution making the quadratic differential a perfect square is $Q(z)dz^2 \mapsto Q(\pi(w))\pi'(w)^2dw^2$ (1.10) where $\pi: \tilde{D} \rightarrow D$ is the double cover.

In either method, for a single branch point one essentially employs the square root transform (cf Remark 2.3). For more branch points and rational quadratic differentials, the covering is elliptic or hyperelliptic.

Nehari's theorems are valid on general Riemann surfaces. It is also possible to generalize our conformal invariants (1.11) and their properties to nested Riemann surfaces (work in progress with Staubach). In this setting the conformal invariants have deep connections with Teichmüller theory.

3.3 Functional differentiation

According to Schiffer's variational methods, the derivative of a reasonable functional on the class of conformal maps is associated with a quadratic differential (Pommerenke 1975). More precisely, the functional derivative, in a given direction, can be expressed as a contour integral pairing between a quadratic differential and holomorphic function tangent to the variation. See for example Roth (1998, 2000), where functional derivatives are obtained through the use of the Loewner and Schiffer variational methods.

One is thus led immediately to the question: what is the functional derivative of $m(D_1, D_2, Q(z)dz^2)$ and its relation to $Q(z)dz^2$? An answer in the case that $Q(z)dz^2 = x'(z)dz^2$ on the disk was obtained by the author in Schippers (2010).

Fix $D_1 = \mathbb{D}$ and $D_2 = f(\mathbb{D})$ for some injective holomorphic map $f: \mathbb{D} \rightarrow \mathbb{D}$ such that $f(0) = 0$. Consider the Nehari functional $M_{\text{Nehari}}(\mathbb{D}, f(\mathbb{D}), \text{Re } x)$ associated with singularity function $\text{Re } x(z)$ for a Laurent polynomial

$$x(z) = \sum_{k=-n}^n x_k(z)z^k.$$

We assume that $x'(z)^2 dz^2$ is a quadratic differential which is admissible for the disk.

Let p_t be a one-parameter family of holomorphic functions of positive real part which is measurable in t , and such that $p_t(0) = 0$, $p'_t(0) = 1$. Let f_t be a solution to the so-called Friedland–Schiffer equation

$$\frac{\partial f_t}{\partial t}(z) = -zp_t(z)f_t(z)$$

on $[0, T]$ with initial condition $f_0(z) = z$. A solution exists by results of Roth (1998).

We then have the following result.

Theorem 3.4 (Schippers 2010) *With f_t , p_t , $x(z)$ and M_{Nehari} as above, if f_{t_0} is an extremal function for M_{Nehari} , then*

$$\left. \frac{d}{dt} \right|_{t=t_0} M_{\text{Nehari}}(\mathbb{D}, f(\mathbb{D}), \text{Re } x) = \lim_{r \nearrow 1} \text{Re} \frac{1}{\pi i} \int_{\gamma_r} Q(z) p_{t_0}(z) \frac{dz}{z},$$

where

$$\frac{Q(z)}{z^2} dz^2 = f_{t_0}^* \left(x'(z)^2 dz^2 \right).$$

Note that extremality implies admissibility for $x'(z)^2 dz^2$. Thus, the functional derivative, at an extremal, reproduces the original quadratic differential. Work in progress with W. Staubach shows that a similar result holds for arbitrary quadratic differentials in a much more general setting.

General formulas for the functional derivative away from extremals were also obtained in Schippers (2010).

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